

NHL Home Ice Advantage

SDS 375

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Topic & Motivation

Examine the presence/significance of home ice advantage in NHL games, along with possible factors that influence it.

It's commonly believed that there is home ice advantage in NHL, and there are many components that may affect home ice advantage. From the strategies used, the audience that drive the spirit of the games, the logistics behind the scenes of each game, to events in the games themselves.

How can this help teams?

Identifying controllable factors that influence home ice advantage can help them develop informed strategies to reduce losses.

Though there have been studies on home ice advantage, we are incorporating less explored factors and examining a range of variables.

Research Question & Hypotheses

To what extent does home ice advantage affect NHL games, and to what extent do travel, rules, and fans contribute to home ice advantage?

RULES - Home teams gain a measurable advantage in NHL games due to benefits from rules such as the last change and favorable face-off positioning.

TRAVEL - Increased travel distance and time zone changes negatively affect visiting teams performance, contributing to home ice advantage.

FANS- Higher crowd attendance and noise levels from home fans enhance player motivation and influence game outcomes in favor of the home team.

Data & Methodology

How we decided to quantify our analysis:

1. **Travel** → Distance traveled by team
2. **Fans** → Attendance at each game
3. **Rules** → Shootout and Faceoff advantages to home team

Our sources included:

1. **Nhlscrapr** - R package: Look at individual games stats
 - i. Unit of obs: One NHL game
 - ii. Key vars: Home team, away team, goals for each team
2. **Deep13**: Quantified distance
 - i. Unit of obs: One NHL team
 - ii. Key vars: physical distance between teams, goals by each team
3. **Hockey Reference**: Analyzed home advantage
 - i. Unit of obs: One game (2020-21 & 2024-25 season)
 - ii. Key vars: home team goals, away team goals, attendance

Data & Methodology

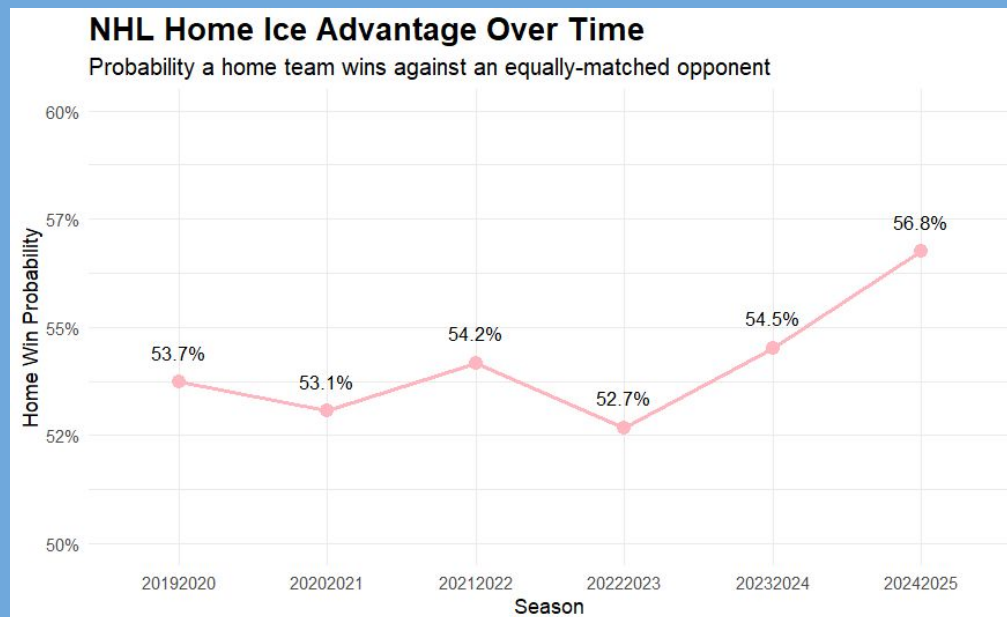
Other sources:

4. Hockey Elo Ratings- csv. File - Faceoff data for any player with 100 faceoff attempts or 300 career attempts in the 2024-2025 season

- i. Unit of obs: One NHL Player faceoff statistics
- ii. Key vars: Name, FP (win percentage when stick down first), LP (win percentage when stick down last).

Results & Interpretation

First, I found evidence which supports that home ice advantage clearly exists in the NHL in each of the past 6 seasons.



season <chr>	intercept_log_odds <dbl>	home_win_probability <dbl>
20192020	0.1496221	0.5373359
20202021	0.1227516	0.5306494
20212022	0.1672231	0.5417086
20222023	0.1071235	0.5267553
20232024	0.1812751	0.5451951
20242025	0.2718929	0.5675575

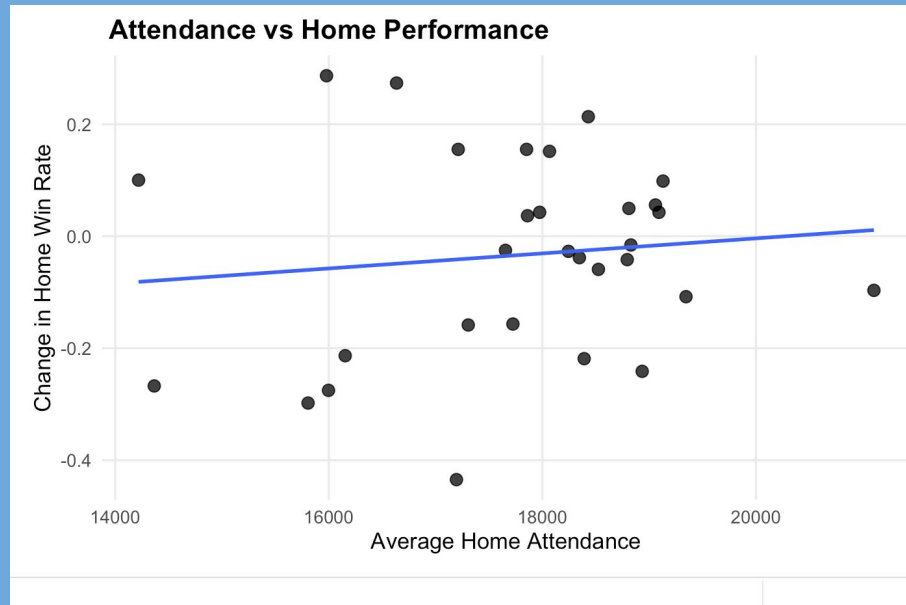
Results & Interpretation

```
# Fit Bradley-Terry model
model <- glm( win ~ Home + Visitor, data = hockey25, family = binomial() )
summary(model)

1/(1+exp(-0.41903))
```

```
Call:
glm(formula = win ~ Home + Visitor, family = binomial(), data = hockey25)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -0.41903    0.47356   -0.885  0.37623
HomeBoston Bruins    0.07469    0.45752    0.163  0.87033
HomeBuffalo Sabres  -0.24521    0.45869   -0.535  0.59294
HomeCalgary Flames  -0.17534    0.45937   -0.382  0.70269
HomeCarolina Hurricanes -1.17974    0.49414  -2.387  0.01697 *
HomeChicago Blackhawks  0.54346    0.46520    1.168  0.24271
HomeColorado Avalanche -0.55471    0.46561   -1.191  0.23351
HomeColumbus Blue Jackets -0.55961    0.46534   -1.203  0.22914
```



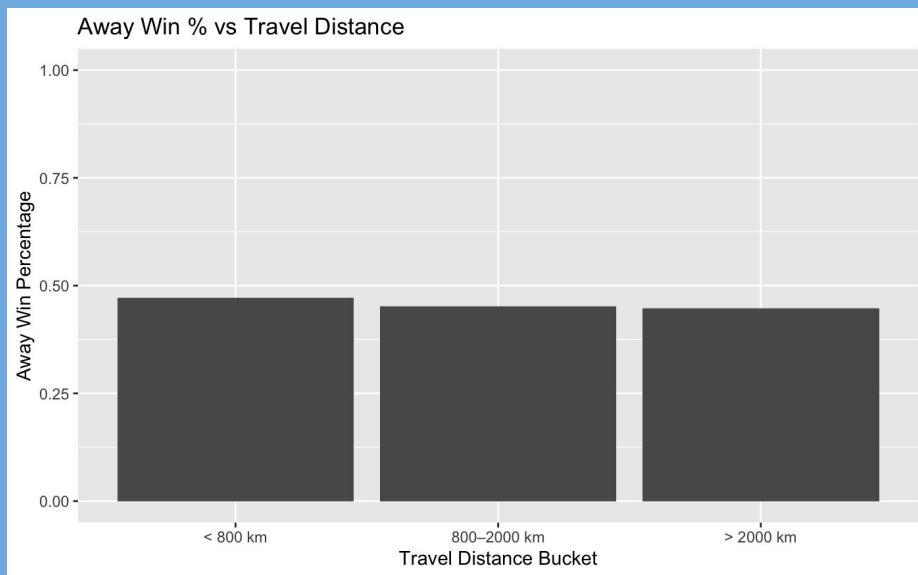
The Bradley Terry intercept of -0.41903 shows that there is a baseline home win probability of about 60.3%. This means that even when two NHL teams are evenly matched, the home team wins approximately 60% of the time when fans are present.

Results & Interpretation

```
Call:
glm(formula = away_win ~ distance_1000km, family = binomial(),
    data = games_travel)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.14716	0.06384	-2.305	0.0212 *
distance_1000km	-0.01585	0.02960	-0.536	0.5923



- Away win percentage only drops slightly as travel distance increases
- Logistic regression shows no statistically significant effect of travel distance on away wins
- Modern travel methods such as charter flights and improved recovery likely reduce fatigue which doesn't hinder players performance substantially
- In conclusion, travel distance isn't a major factor behind home ice advantage in recent NHL seasons

Results & Interpretation

- In the NHL it is a disadvantage to put your stick down first on faceoffs
- The visiting team puts their stick down first at center ice (otherwise it is the defensive team)

- After analyzing a data set of 193 NHL players during the 2024-2025 season, it is found that there is a 1.542% increase in faceoff win percentage putting your stick down last as opposed to first

Average Change (Last vs First): 1.542 %

Conclusion & Future Direction

- Based on attendance, the home team is more likely to win when evenly matched.
- Travel distance only has a very slight negative effect on away team performance - not enough evidence for it to be a significant factor.
- Stick down first does offer a slight advantage in faceoff wins.

Further research:

- Examine home ice advantage over multiple seasons so see if trends stay consistent.
- Try nonlinear models, ML prediction models.
- Take into account any major changes in rules or other circumstances.

Thank You